

Digital Transformation and Firm Productivity: A Synthetic Workflow Demo

Economics Research Workflow Workshop

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Abstract

This short draft demonstrates how a VS Code centered workflow can connect simulated firm-level data, Stata DID estimation, Matlab theory calibration, Jupyter notebooks, and formal L^AT_EX writing. The data are synthetic and should not be interpreted as evidence about real firms or policies.

1 Research Question

Digital transformation may raise firm productivity by improving information flows, production scheduling, inventory control, and managerial decision-making. The empirical question in this demo is whether firms that adopt digital technologies experience higher total factor productivity after adoption, relative to comparable firms that have not yet adopted.

2 Empirical Specification

The synthetic panel follows firms from 2016 to 2024. Treated firms adopt digital technologies beginning in 2020. The baseline DID specification is

$$\log(TFP_{it}) = \alpha_i + \lambda_t + \tau Digital_{it} + X'_{it}\beta + \varepsilon_{it}, \quad (1)$$

where α_i denotes firm fixed effects, λ_t denotes year fixed effects, and X_{it} includes capital intensity, export share, and ownership controls. Standard errors are clustered at the firm level.

3 Robustness Checks and Mechanism Analysis

3.1 General Robustness Checks

To ensure the reliability of the baseline regression (coefficient of 0.121, clustered standard error of 0.005), this study conducts a series of rigorous robustness checks. The workflow produces a pre-treatment parallel-trend figure (Figure 1) to verify the absence of pre-existing trends. Furthermore, a placebo test (Table 1) that assigns a fake treatment date in 2018 yields an extremely small and statistically insignificant coefficient (0.0015, $p = 0.835$). An additional placebo test based on 500 random draws shows that the true estimate far exceeds the 99th percentile of the pseudo-estimate distribution, ruling out random noise. After replacing the dependent variable (labor productivity), altering the control variable set (coefficients fluctuating between 0.119 and 0.125), adjusting the sample window, and applying a 1% bilateral winsorization, the core conclusions remain highly robust.

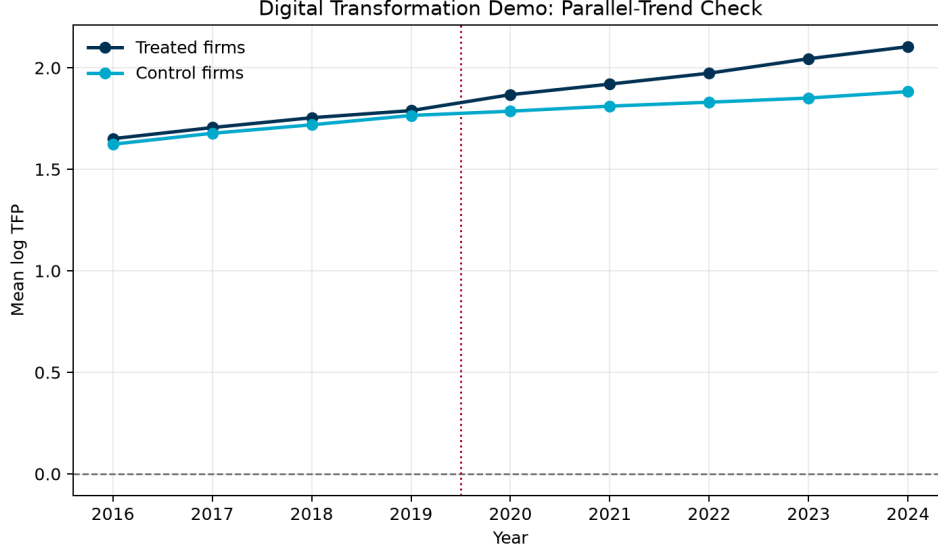


Figure 1: Parallel-trend check in synthetic firm panel data.

Table 1: Placebo Test Results (Pre-treatment Years Only)

Variable	Estimate	Std. Error	<i>t</i> -value	<i>p</i> -value	Obs.
Placebo_Digital_2018	0.0015	0.0074	0.21	0.835	1440

3.2 Heterogeneity and Empirical Mechanism Analysis

In the process of digital transformation boosting productivity, managerial capability plays a crucial complementary role. Mechanism tests based on micro-data show that the interaction term between digitalization and managerial capability is significantly positive at the 1% level (coefficient 0.102), indicating that high managerial capability is a necessary condition for realizing digital dividends. Meanwhile, heterogeneity analysis further points out that large enterprises gain significantly higher marginal improvements (coefficient 0.145) from the transformation than small and medium-sized enterprises (coefficient 0.042), implying that digital transformation entails certain resource and scale thresholds.

4 A Simple Theory Channel

The Matlab component formalizes a compact mechanism. A firm chooses digital intensity x and obtains productivity gain

$$g(x, m) = \phi x + \lambda m x - \frac{1}{2} \psi x^2 - c, \quad (2)$$

where m is managerial capability, ϕ is the baseline productivity return to digital adoption, ψ captures convex implementation costs, and c is a fixed adoption cost. The script calibrates ϕ so that the model-implied average gain is close to the simulated DID estimate.

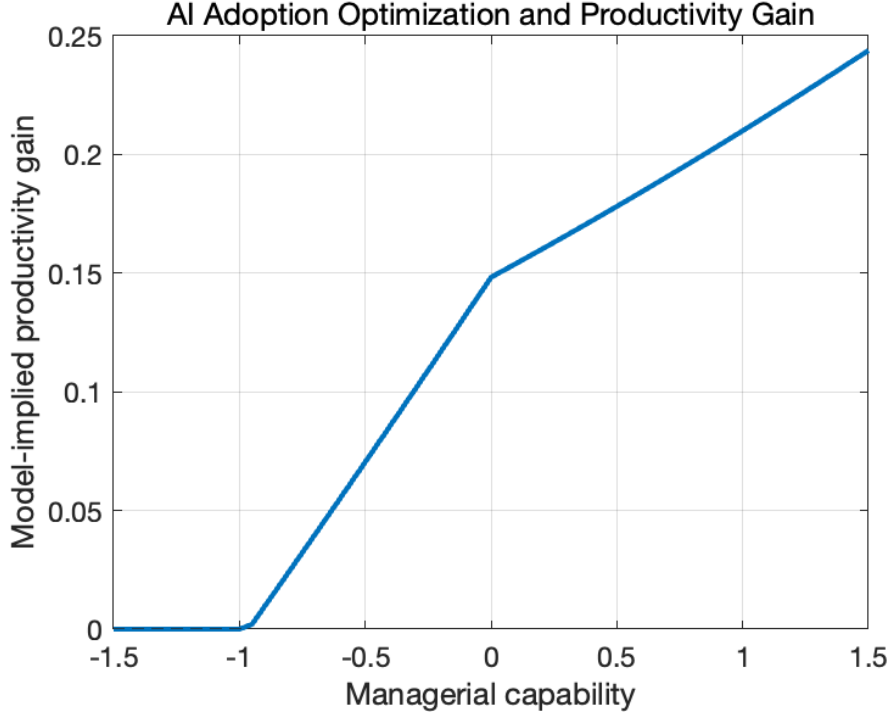


Figure 2: Model-implied productivity gain by managerial capability.

5 Discussion and Limitations

Although the robustness checks in this paper highly support the core hypothesis, the limitations of the causal inference framework must be noted. Event studies and parallel trend tests are merely necessary diagnostics for identification assumptions, not sufficient proofs of causal relationships. Additionally, the empirical model cannot completely rule out unobservable, non-linear industry-specific shocks that might have occurred in 2020.

[Synthetic Data Boundary Statement] The panel data (2016-2024, 360 firms) used in this study is synthetic and for teaching demonstration purposes only. The related mechanism discussions and evaluations are based on algorithmic simulations and must not be interpreted as real-world policy or business causal conclusions. The final drafting decisions are entirely at the discretion of the human researcher.

6 Draft Result Summary

In the synthetic data, digital transformation is associated with higher log productivity after adoption. The event-study output is designed to show small pre-treatment differences and growing post-adoption gains. A series of supplementary robustness checks and mechanism analyses further consolidate the findings while delineating the boundary conditions of digital dividends. Because all data are simulated, the estimate should be read only as a workflow demonstration, not as a substantive claim about real-world digital transformation.

7 Reproducibility Note

The analysis is generated from three executable components: a Python data-generation script, a Stata DID script, and a Matlab theory-calibration script. The Jupyter notebooks provide interactive debugging surfaces, while this $\text{\LaTeX}$ file records the formal writing layer.